

ACE: A Novel and Biologically Inspired 5-Factor SNN Training Rule

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Abstract—The current standard biologically plausible training rule for spiking network reinforcement learning (RL), reward-modulated spike-timing-dependent plasticity (RSTDP), often has inferior performance compared to biologically implausible methods. In addition, RSTDP is slow, purely local, and fragile in non-stationary environments. This paper presents ACE (the first three letters of acetylcholine), a novel, biologically inspired learning rule that extends RSTDP’s three factors to five. ACE introduces modulating factors based on neuromodulators acetylcholine (ACh) and norepinephrine (NE), which modulate learning rate and random noise, respectively. Actor-critic systems trained using ACE achieve higher reward over the fixed window following a task switch than a hyperparameter-tuned classic RSTDP baseline. On the Switch Bandit tasks, ACE recovers more quickly than the baseline; on Switch CartPole, the classic baseline fails to re-adapt within the training horizon while ACE recovers control. ACE does not match non-local surrogate-gradient training but still outperforms classic RSTDP in the tested benchmarks. ACE demonstrates that biologically-inspired neuromodulation increases the adaptability of SNNs in the tested non-stationary environments.

Index Terms—spiking neural networks, reinforcement learning, neuromodulation

I. INTRODUCTION

Spiking neural networks (SNNs) are neural networks that encode information using discrete spike events rather than continuous-valued activations, as used by classical artificial neural networks (ANNs) [1]. Due to their event-driven nature, SNNs can perform much more efficiently, particularly when used on neuromorphic computing hardware [2]. Since computation occurs only when spike events are generated, SNNs avoid the dense activation of classical ANNs [3].

Typical training methods fall in three different categories. Surrogate gradients convert the spikes of SNNs into continuous, differentiable gradients, allowing backpropagation to run [4]. Another training method deploys an ANN-SNN conversion, mapping the weights of a backpropagation-trained ANN to an SNN’s spiking rates [5]. Both of these methods are quite effective, but neither are biologically plausible [6]. Biological brains do not use gradients to strengthen/weaken synapses, and they definitely do not train and convert from ANNs. A more biologically plausible method is spike-timing-dependent plasticity (STDP) [7], which measures the relative timings between neurons to adjust weights, inspired by biological Hebbian learning processes [8], [10]. Expanding STDP to RL, Razvan V. Florian further developed reward-modulated spike-timing-dependent plasticity (RSTDP), mimicking biological

dopamine [9], [11]. A limitation of classical RSTDP is its fragility in non-stationary environments, encountered in many live deployment scenarios. Its reliance on a single modulatory signal can lead to degraded performance when task dynamics change. Biological learning, by contrast, is shaped by the interaction of multiple neuromodulatory systems. RSTDP therefore captures only a narrow subset of the mechanisms underlying biological learning, motivating extended learning rules that incorporate additional neuromodulatory factors.

To investigate whether additional neuromodulatory signals can improve RSTDP-style adaptation under task switches, this paper introduces ACE, an extension of RSTDP that incorporates modulation of the learning rate and injected noise levels, emulating biological acetylcholine (ACh) and norepinephrine (NE), respectively. Specifically, ACh adaptively scales synaptic plasticity to enable rapid re-learning when the environment changes, while NE adaptively scales stochasticity in action selection to encourage exploration during periods of uncertainty. ACh and NE each depend on an abstracted expected uncertainty and unexpected uncertainty, respectively, as outlined by Yu and Dayan [17]. ACE consistently improves post-switch adaptation relative to baseline RSTDP, supporting the hypothesis neuromodulation improves the adaptation of shallow spiking neural networks in non-stationary reinforcement learning tasks. In addition, we provide ablations and statistical testing to reduce the likelihood that gains are attributable to tuning artifacts.

II. RELATED WORK

Recently, many works have sought after biological plausibility with their learning rules. In 2020, Bellec et al. introduced Eligibility Propagation (e-prop), which uses eligibility traces with a learning signal to adjust connections [12]. However, e-prop still relies on online approximations of loss gradients, which is not too biologically plausible. Further, Deep Continual Local Learning (DECOLLE), introduced by Kaiser, Mostafa, and Neftci in 2020, uses synthetic gradients in a more local manner than classical backpropagation [13]. Very recently, Tihomirov et al. [14] deployed an actor-critic system on RL benchmarks cartpole and acrobot, training the actor on RSTDP and the critic on temporal difference long-term potentiation (TD-LTP), a similar local learning rule to RSTDP; though they used less biologically plausible Ising neuron models, they demonstrate that RSTDP can solve both reliably.

Beyond a single global reward signal, recent work has explored more robust modulatory structures for credit assignment in SNNs. Liu et al. (2021) introduced a framework where modulatory signals improve how recurrent spiking networks assign credit over time [15]. Schmidgall and Hays (2023) introduced Meta-SpikePropamine, where neuromodulated plasticity rules are learned within a meta-learning loop [16]; however, it relies heavily on gradient descent, unlike ACE.

Doya [20] provides the foundational meta-learning framework underlying this work, proposing that distinct neuromodulators each control a separate reinforcement learning meta-parameter: dopamine signaling reward prediction error, acetylcholine controlling learning rate, and norepinephrine controlling the inverse temperature of action selection. ACE utilizes the ACh and NE roles from this framework within a spiking actor-critic. Later, Yu and Dayan [17] further conjectured that ACh tracks expected uncertainty, while NE tracks unexpected uncertainty, which can trigger rapid changes in inference and learning. Complementing this, Aston-Jones and Cohen proposed the LC-NE adaptive gain theory in 2005, linking NE dynamics to shifts in exploration and exploitation [18], and Belkaid and Krichmar (2020) proposed a model in which cholinergic modulation of dopaminergic activity drives uncertainty-seeking behavior [19]. ACE follows Yu and Dayan’s work, utilizing abstractions of ACh and NE derived from critic variance (the expected uncertainty) and TD-error (the unexpected uncertainty), respectively.

III. THE ACE SYSTEM

A. Network Architecture

The ACE training system was deployed on an actor-critic system. Each neuron’s state in the actor is defined by the following equations:

$$\begin{aligned} z_i[t] &= \lambda z_i[t-1] + x_i[t], & \lambda &= e^{-\Delta t/\tau_m} \\ v_k[t] &= \sum_i W_{ik} z_i[t] + \xi_k[t] \\ p_k[t] &= 1 - \exp\left(-\rho_0 \exp\left(\frac{v_k[t] - v_{th}}{\kappa}\right) \Delta t\right). \end{aligned} \quad (1)$$

In Eq. 1, each input neuron i emits a binary spike $x_i[t] \in \{0, 1\}$ at timestep t , which is integrated by the leaky trace $z_i[t]$, where Δt is the step size and τ_m the membrane time constant. The noisy membrane drive $v_k[t]$ of output unit k combines these traces through the synaptic weights W_{ik} from input neuron i to output unit k , plus an additive noise term $\xi_k[t]$ governed by NE. In the instantaneous spiking probability $p_k[t]$, ρ_0 sets the baseline firing-rate scale, v_{th} is the firing membrane threshold, and κ controls the steepness with which firing probability increases as the membrane potential exceeds threshold. Output spikes are sampled stochastically according to $p_k[t]$ and mapped to a discrete action. When multiple output units spike or no output unit spikes within a timestep, the action is instead selected as $\arg \max_k v_k[t]$, ensuring a unique action is chosen at each timestep. The actor is a single-layer spiking policy without a hidden layer. The critic consists of two linear readouts operating directly on the input spikes: a value readout producing $V(s_t)$ and a variance readout

producing $\sigma^2(s_t)$, with a softplus nonlinearity applied to the variance head to ensure positivity.

B. Benchmarks

1) *Switch CartPole*: Switch CartPole is a modified version of the standard CartPole task in which the controls of the agent are inverted halfway through training. The continuous CartPole state $s_t = (x, \dot{x}, \theta, \dot{\theta})$ is converted into a population-based spiking representation using fixed Gaussian place cells. Place-cell centers are arranged on a Cartesian grid over the state space, and each cell produces spikes according to a Poisson process with rate

$$r_i(s_t) = \rho_{PC} \exp\left(-\sum_d \frac{(s_{t,d} - \mu_{i,d})^2}{2\sigma_d^2}\right),$$

where μ_i is the preferred state of neuron i and σ_d is set from the grid spacing along dimension d (Table I). Rates are converted to per-timestep spike probabilities and sampled independently.

2) *Switch Bandit*: Switch Bandit is a minimal non-stationary multi-armed bandit task where the agent selects an arm and gradually learns the probability with which each arm returns a reward. In our implementation of the default switch bandit, the agent selects one of two actions at each episode; selecting the optimal arm yields a guaranteed reward, selecting a suboptimal arm yields a guaranteed punishment, and the probabilities of each arm are swapped at the switch instantaneously (Table I). This task removes state dynamics, isolating the agent’s ability to adapt to changes in reward trends. The Multi-switch Bandit follows the same switching regime over four arms rather than two, with the correct arm inverted at episodes 15,000, 30,000, and 45,000 of a 60,000-episode run. Gradual Switch Bandit swaps the two arms’ probabilities by linearly annealing each toward the other’s value over a given number of episodes (200 in our implementation).

C. Training Methods and Neuromodulation

1) *Surprise*: ACE bases its neuromodulation on two surprise signals corresponding to expected and unexpected uncertainty, following the framework of Yu and Dayan [17]. The unexpected uncertainty signal is derived from the novelty of the temporal-difference (TD) error. At each timestep, we define a TD-derived signal $u_t = |\delta_t|$. This signal is passed through two exponential moving averages with different timescales:

$$\begin{aligned} u_t^f &= (1 - \alpha_f) u_{t-1}^f + \alpha_f u_t, \\ u_t^s &= (1 - \alpha_s) u_{t-1}^s + \alpha_s u_t, \end{aligned} \quad (2)$$

where u_t^f is a fast trace capturing recent deviations and u_t^s is a slow trace representing the long-term baseline. Unexpected surprise is defined as the rectified difference between the fast and slow traces,

$$n_{u,t} = \max(0, u_t^f - u_t^s - m), \quad (3)$$

where m is a small dead-zone margin that suppresses sensitivity to minor fluctuations. To obtain a stable control signal for neuromodulation, this novelty measure is accumulated using a leaky integrator,

$$\bar{n}_{u,t} = \beta_u \bar{n}_{u,t-1} + n_{u,t}, \quad (4)$$

with $\beta_u < 1$. The leaky integration in Eq. 4 acts as a temporal smoothing filter, mirroring the sustained release and clearance times of biological neuromodulators. The expected uncertainty signal is derived from the critic’s predicted standard deviation $\sigma(s_t) = \sqrt{\sigma^2(s_t)}$, which captures the noise within the current task context. We define $n_{e,t} = \sigma(s_t)$, and smooth it with an exponential moving average to obtain a stable control signal,

$$\bar{n}_{e,t} = (1 - \beta_e) \bar{n}_{e,t-1} + \beta_e n_{e,t}. \quad (5)$$

Unlike the unexpected pathway, the expected pathway uses the critic’s variance head directly without a fast-slow novelty decomposition, since the variance estimate already captures within-context noise. In the Switch Bandit implementation, $n_{e,t}$ is set to the predicted variance $\sigma^2(s_t)$ rather than its square root; we found this scaling to interact better with the bandit’s coarser surprise dynamics. In our implementation, u_t is clipped to a fixed maximum for stability, and the fast/slow traces are initialized on the first timestep by setting $u_t^f = u_t^s = u_t$ (Table I).

2) *Actor Training*: The actor trains according to an extended RSTDTP defined like so:

$$\Delta W_{ik} = \eta_t \delta_t z_i[t] S_k[t], \quad (6)$$

where η_t is a leaky-integrated learning rate driven by the ACh signal:

$$\eta_t = \text{clip}((1 - \lambda_\eta) \eta_{t-1} + \kappa_\eta \text{ACh}_t, \eta_{\min}, \eta_{\max}), \quad (7)$$

and the ACh signal itself is a logistic function of the smoothed expected uncertainty:

$$\text{ACh}_t = \frac{\eta_{\max}^{\text{ACh}}}{1 + \exp(-k_{\text{ACh}}(\bar{n}_{e,t} - c_{\text{ACh}}))} + \eta_{\text{base}}. \quad (8)$$

In Eq. 6, $z_i[t]$ is the presynaptic eligibility trace (Eq. 1) and $S_k[t] \in \{0, 1\}$ is the postsynaptic spike of output neuron k . In Eq. 7, λ_η is the decay rate and κ_η the boost coefficient, with η_t bounded between $\eta_{\min}$ and $\eta_{\max}$; this integration smooths the effective plasticity rate and prevents abrupt changes from individual ACh spikes. In Eq. 8, $\eta_{\max}^{\text{ACh}}$, k_{ACh} , and c_{ACh} are the maximum ACh contribution, the steepness of the logistic, and its activation center, and η_{base} is a small baseline ensuring nonzero ACh in the absence of expected uncertainty. For stability, synaptic weights are maintained within a bounded range. In the Switch Bandit implementation, the leaky-integrator dynamics (Eq. 7) are disabled, and η_t is set directly to ACh_t at each step, as each episode is a single decision/timestep.

3) *Norepinephrine Modulation*: NE is computed as a logistic function of the smoothed unexpected uncertainty signal $\bar{n}_{u,t}$, following the distinction of Yu and Dayan [17] in which ACh tracks expected uncertainty within a known context while NE tracks unexpected change in task dynamics. Elevated NE increases additive noise, promoting exploration when the agent encounters novel conditions; as unexpected uncertainty diminishes, NE levels decrease, reducing noise and allowing behavior to stabilize around the learned policy. We set the noise standard deviation as

$$\sigma_{\text{NE}}(\bar{n}_{u,t}) = \frac{\sigma_{\max}}{1 + \exp(-k_{\text{NE}}(\bar{n}_{u,t} - c_{\text{NE}}))} + \sigma_0, \quad (9)$$

and sample $\xi_k[t] \sim \mathcal{N}(0, \sigma_{\text{NE}}(\bar{n}_{u,t})^2)$ independently for each output unit and timestep (with σ_{NE} clipped to a fixed maximum in implementation). Here $\sigma_{\max}$, k_{NE} , c_{NE} , and σ_0 are the maximum NE-driven noise contribution, the steepness, the activation center, and a small baseline noise term, mirroring the ACh mapping. The sigmoidal shape prevents runaway noise while keeping the system responsive to meaningful task changes.

4) *Critic Training*: The critic is trained at each timestep using a local TD-LTP rule. A temporal-difference (TD) target is computed as

$$y_t = r_t + \gamma V(s_{t+1}), \quad (10)$$

$$\delta_t = y_t - V(s_t),$$

where r_t is the reward and γ is the discount factor; for terminal states, $V(s_{t+1})$ is set to zero, and δ_t is the resulting TD error.

The value and variance weights are updated using local plasticity rules of the form

$$\Delta w_{\text{val}} = \alpha_c |\delta_t| \delta_t x_t, \quad (11)$$

$$\Delta w_{\text{var}} = \alpha_c (\delta_t^2 - \sigma^2(s_t)) x_t,$$

where α_c is the critic learning rate and x_t is the input spike vector. The value rule scales the update by the squared TD error magnitude $|\delta_t|^2$ while preserving the sign of δ_t , providing a TD-LTP-style update in which large surprises drive proportionally larger changes. The variance rule drives $\sigma^2(s_t)$ toward the squared TD error δ_t^2 . Bias terms are updated analogously without the input spike modulation.

For Switch Bandit, the same critic structure is used, but the task is treated as a contextual bandit with no state transitions. The TD target reduces to the immediate reward, $y = r$, and the TD error is $\delta = r - V(s)$.

5) *Training Procedure and Reproducibility*: Network parameters were initialized randomly, and no parameters or neuromodulatory states were reset at the environment switch point. Training horizons and switch points are reported per-task in Table I. Training was executed on a CPU on an Apple M1 system, without GPU acceleration.

IV. RESULTS

ACE was evaluated on the aforementioned benchmarks and compared to the same system trained with classic RSTDTP (with neuromodulation turned off). For the classic system on Switch Cartpole, the learning rate was set to 5×10^{-4} and the base noise to 0.5; on Default Switch Bandit, the learning rate was set to 10^{-2} . We further ablated the individual functions of ACh and NE, evaluating ACE with each held constant in turn, and provide hyperparameter sensitivity analyses for both ACE and Classic. Additionally, a surrogate baseline is provided for Cartpole, as it is the more complex control task. It uses the same place-cell encoder as ACE, driving a single linear layer into a leaky integrate-and-fire (LIF) output population (one unit per action), implemented via `snnTorch` with the fast-sigmoid surrogate gradient (slope = 25). Both networks are trained with REINFORCE using Adam and gradient clipping (max norm 1.0); the actor with the standard policy-gradient loss plus entropy regularization (learning rate

TABLE I
TASK-SPECIFIC HYPERPARAMETERS AND IMPLEMENTATION DETAILS.

Parameter	Switch CartPole	Switch Bandit
Episode count / switch episode	10,000 / 5000	20,000 / 10,000
Shared by both tasks: $\Delta t = \tau_m = 0.02$; $v_{th} = \kappa = 2.0$; $\rho_0 = 100$; u_t clip = 20; novelty margin $m = 0$; unexpected leak $\beta_u = 0.8$		
Discount factor γ	0.99	–
Place-cell peak rate ρ_{PC}	50	–
Place-cell grid centers μ_i	x : 6 in $[-2.5, 2.5]$;	–
	$\dot{x}$: 6 in $[-2, 2]$;	
	θ : 8 in $[-0.25, 0.25]$;	
	$\dot{\theta}$: 8 in $[-2, 2]$	
# place cells (input neurons)	$6 \times 6 \times 8 \times 8 = 2304$	–
σ_d from grid spacing	spacing/1.5	–
Fast/slow trace rates (α_f, α_s)	(0.0977, 0.00364)	(0.0977, 0.1)
Expected EMA β_e (Eq. 5)	0.01	1.0
ACh: $\eta_{max}^{ACh}, k_{ACh}, c_{ACh}, \eta_{base}$	1.0, 10, 5, 1×10^{-4}	1.0, 5, 1.5, 10^{-2}
NE: $\sigma_{max}, k_{NE}, c_{NE}, \sigma_0$ (Eq. 9)	3, 0.2, 15, 1	2, 1.0, 1.5, 0
Actor LR dynamics: $\lambda_\eta, \kappa_\eta, \eta_{min}, \eta_{max}$ (Eq. 7)	0.1, 0.1, $10^{-4}, 0.1$	– (disabled)
Critic architecture	linear value + linear variance (softplus)	linear value + linear variance (softplus)
Critic learning rate α_c	10^{-3}	10^{-2}

3×10^{-3} , entropy coefficient 1×10^{-3} , action temperature 1.0), the critic with mean squared TD error (learning rate 1×10^{-3}). Each condition was evaluated across 20 seeds. All comparisons below are paired across these seeds, and we report two-sided p -values from a paired t -test and a Wilcoxon signed-rank test alongside the mean paired difference. With 20 paired seeds the exact Wilcoxon minimum is 1.9×10^{-6} ; that value therefore indicates ACE won in every seed.

A. Switch Bandit

TABLE II
SWITCH BANDIT RESULTS (20 PAIRED SEEDS PER BENCHMARK).
MEAN $\pm$ STANDARD DEVIATION (SD) ACROSS SEEDS FOR CLASSIC
RSTDP (NEUROMODULATION OFF) AND ACE. TOTAL REWARD IS
ACCUMULATED OVER THE FULL RUN; POST-SWITCH REWARD IS
ACCUMULATED AFTER THE SWITCH.

Benchmark	Window	Classic RSTDP	ACE
Default	Total	7,885.8 $\pm$ 1,976.8	14,667.5 $\pm$ 1,416.2
Default	Post-switch	–1,526.0 $\pm$ 2,049.4	4,746.2 $\pm$ 1,411.4
Gradual	Total	7,491.1 $\pm$ 2,190.4	14,298.9 $\pm$ 1,252.1
Gradual	Post-switch	–1,920.7 $\pm$ 2,286.2	4,377.6 $\pm$ 1,247.6
Multi-switch	Total	–211.3 $\pm$ 1,092.8	1,530.5 $\pm$ 1,933.7

On each Switch Bandit task, ACE outperformed its classic counterpart. On the default benchmark, ACE achieved a mean paired improvement of +6781.7 in total reward across 20 seeds (paired t -test $p = 2.1 \times 10^{-9}$; Wilcoxon signed-rank $p = 1.9 \times 10^{-6}$), and the gap widened after the switch, where ACE outperformed classic RSTDP in all 20 seeds for a mean paired improvement of +6,272.2 (paired t -test $p = 1.1 \times 10^{-8}$; Wilcoxon $p = 1.9 \times 10^{-6}$; Table II). The gradual variant similarly had mean paired improvements of +6,807.8 in total reward (paired t -test $p = 2.7 \times 10^{-9}$; Wilcoxon $p = 1.9 \times 10^{-6}$) and +6,298.3 post-switch (paired t -test $p = 1.6 \times 10^{-8}$; Wilcoxon $p = 1.9 \times 10^{-6}$). Under repeated reversals, ACE achieved a mean paired improvement of +1,741.8 (paired t -test $p = 0.014$; Wilcoxon $p = 0.021$).

B. Switch CartPole

On the Switch CartPole task, ACE demonstrated a clear advantage in adaptation following the environment switch. Across 20 paired seeded runs, ACE consistently recovered more rapidly after the switch.

TABLE III
SWITCH CARTPOLE: MEAN PER-EPISODE REWARD (20 PAIRED
SEEDS). MEAN $\pm$ SD ACROSS SEEDS FOR CLASSIC RSTDP, ACE, AND
THE SURROGATE-GRADIENT BASELINE.

Method	Full run	Post-switch
Classic RSTDP	88.71 $\pm$ 13.52	9.825 $\pm$ 0.0380
ACE	148.8 $\pm$ 31.91	172.2 $\pm$ 58.77
Surrogate	340.0 $\pm$ 61.27	248.4 $\pm$ 25.60

As shown in Table III, ACE achieved a mean paired improvement of +60.09 in per-episode reward (paired t -test $p = 2.0 \times 10^{-8}$; Wilcoxon $p = 1.9 \times 10^{-6}$). To isolate adaptation performance, Table III reports per-episode reward after the dynamics switch. ACE yielded a mean paired improvement of +162.4 across 20 seeds (paired t -test $p = 1.6 \times 10^{-10}$; Wilcoxon $p = 1.9 \times 10^{-6}$). While Classic RSTDP achieves stable performance prior to the switch, it adapts poorly afterward. In contrast, ACE recovers much more reward following the switch. We additionally compared ACE against the surrogate-gradient baseline described above (Tables III and III). On full-run reward, the surrogate baseline outperformed ACE, a difference significant under a paired t -test (paired t -test $p = 1.5 \times 10^{-14}$). The gap narrowed in the post-switch regime (paired t -test $p = 4.5 \times 10^{-3}$).

C. Ablation Study

TABLE IV
ABLATIONS: TOTAL REWARD (20 PAIRED SEEDS). MEAN $\pm$ SD ACROSS
SEEDS FOR FULL ACE AND THE TWO REDUCED VARIANTS, ON SWITCH
BANDIT AND SWITCH CARTPOLE.

Variant	Switch Bandit	Switch CartPole
Full ACE (ACh + NE)	14,667.5 $\pm$ 1,416.2	148.8 $\pm$ 31.91
ACh-only (NE constant)	6,290.7 $\pm$ 1,535.7	138.4 $\pm$ 21.25
NE-only (ACh constant)	14,048.8 $\pm$ 1,259.1	67.96 $\pm$ 9.058

1) *Switch Bandit Ablation*: On the Switch Bandit task, removing ACh modulation yielded a small mean paired decrement of –618.7 across 20 seeds (paired t -test $p = 0.038$; Wilcoxon $p = 0.040$) (Table IV). Removing NE modulation yielded a significantly larger mean decrement of –8,376.8 (paired t -test $p = 5.0 \times 10^{-12}$; Wilcoxon $p = 1.9 \times 10^{-6}$).

2) *Switch Cartpole Ablation*: On the Switch CartPole task, the contributions of the two modulators reversed (Table IV). Removing NE modulation (ACh-only variant) had no significant effect, with the paired decrement of –10.4 not significant (paired t -test $p = 0.22$; Wilcoxon $p = 0.25$). In contrast, removing ACh modulation significantly degraded performance, a mean paired decrement of –80.8 across 20 seeds. This difference was highly significant (paired t -test $p = 1.3 \times 10^{-10}$; Wilcoxon $p = 1.9 \times 10^{-6}$).

D. Hyperparameter Sweeps

ACE sensitivity. Sampling 100 configurations from a nine-parameter grid confirms that ACE is genuinely hyperparameter sensitive: total per-episode reward spanned 22.4 to 186.3 (Table V), and the average sampled configuration fell below the classic on total reward, with only better-tuned configurations exceeding it. Post-switch, however, the mean across all 100 configurations exceeded the classic’s roughly fivefold.

Baseline fairness. A 5×5 grid over the classic model’s two most influential free parameters found no configuration that recovers post-switch: reward stayed at failure (Table V), confirming the validity of the Classic baseline.

TABLE V

HYPERPARAMETER SWEEPS ON SWITCH CARTPOLE. MEAN PER-EPISODE REWARD ($\pm$ SD). *ACE*: 100 CONFIGURATIONS DRAWN BY RANDOM SAMPLING WITHOUT REPLACEMENT FROM A NINE-PARAMETER GRID OF 1,728 (MINIMUM ACTOR LEARNING RATE, BASELINE EXPLORATION NOISE, AND THE MAXIMUM GAIN, STEEPNESS, AND CENTER OF BOTH THE NE AND ACH DRIVES, PLUS THE CRITIC BASE LEARNING RATE, EACH AT TWO TO THREE LEVELS). *Classic*: 5×5 GRID OVER $\eta \in \{10^{-4}, 3 \times 10^{-4}, 5 \times 10^{-4}, 10^{-3}, 3 \times 10^{-3}\}$ AND $\sigma \in \{0.5, 1.0, 1.5, 2.0, 3.0\}$; ALL CONFIGURATIONS WERE TRAINED OVER 3 SEEDS. THE TOP-RANKED CONFIGURATIONS WERE RE-EVALUATED OVER 5 SEEDS. THE ORIGINAL CLASSIC BASELINE IS INCLUDED.

Configuration set	Total	Post-switch
ACE sweep (100 configs)	62.3 ± 42.4	51.2 ± 62.6
ACE top-10 revalidation	106.7 ± 61.1	153.8 ± 96.2
Classic sweep (25 configs)	52.2 ± 25.1	≈ 11
Classic top-5 revalidation	84.0 ± 9.2	≈ 11
Classic baseline (main)	88.7 ± 13.5	9.8 ± 0.1

V. DISCUSSION AND CONCLUSION

This paper presents ACE, an SNN training method that utilizes biologically inspired neuromodulation. Its superior performance in non-stationary environments relative to classic RSTDp was demonstrated using the switch benchmarks, and the ablations demonstrated that different tasks may require different neuromodulation dynamics. This divergence follows from the differing structural demands of the two task spaces. Switch Bandit is a static decision problem with no state transitions: when the choices flip, an agent without noise continues to exploit its existing policy, failing to adapt. Switch CartPole, by contrast, is a continuous, unstable feedback-control task in which undirected exploration noise is actively destabilizing, so the system must instead rapidly overwrite its original policy, which is what ACh provides. The optimal balance between ACh and NE is task-dependent. Despite these encouraging results, several limitations of the current implementation remain to be addressed in future work. First, the actor network in our experiments is a single-layer spiking architecture. More complex environments will likely require more complex architectures. Second, the critic, since it is a single linear readout and relies on an explicit variance readout $\sigma^2(s_t)$ and squared TD errors δ_t^2 , represents a functional abstraction. Future work could integrate a fully biologically plausible critic. Third, the non-stationary tasks used in this study featured immediate or minimally delayed feedback.

Future work could seek integrating ACE with dedicated credit assignment mechanisms.

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